

Mixed Comparison Report:
CNN ReLU vs. SNN (Atan, Tri, Box) vs. CNN Clipped vs. CNN Tanh

ByzFL SNN Benchmark

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1 Introduction

This report compares six configurations:

1. **CNN ReLU** ($LR = 0.15$): Standard CNN with ReLU, no clipping.
2. **SNN Atan** ($\alpha = 1.2$): Spiking CNN with ArcTangent surrogate.
3. **SNN Tri** ($\beta = 2.0$): Spiking CNN with Triangular surrogate.
4. **SNN Box** ($\beta = 2.0$): Spiking CNN with Box surrogate.
5. **CNN ReLU (Grad Clipped)**: CNN with explicit gradient norm clipping.
6. **CNN Tanh**: CNN with Tanh activation (implicit saturation).

Note: CNN Clipped heatmaps are available for best-aggregator and Geometric Median sections only. Other per-aggregator sections show 5 configs (without Clipped).

2 Best Overall Performance (Worst-Case Across Attacks)

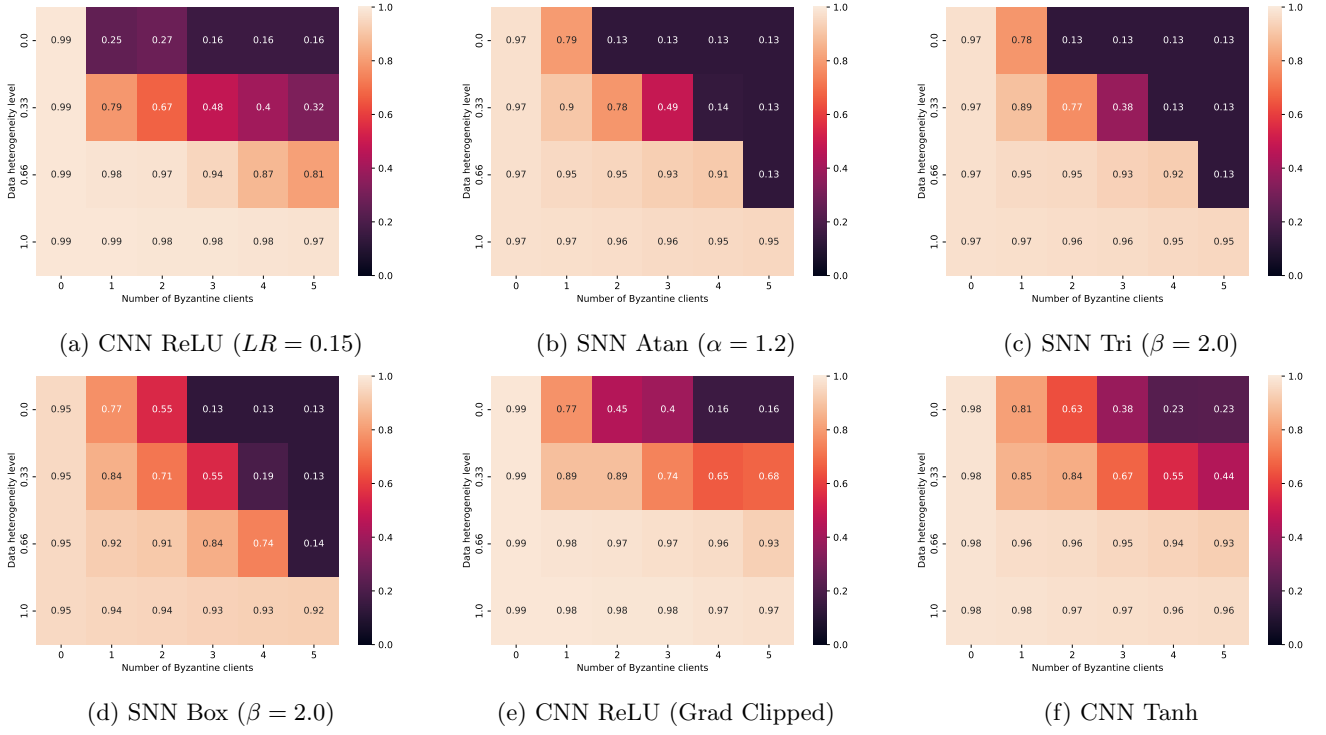


Figure 1: Best overall test accuracy (worst-case across attacks).

3 Best Aggregator under Specific Attacks

3.1 Optimal ALIE Attack

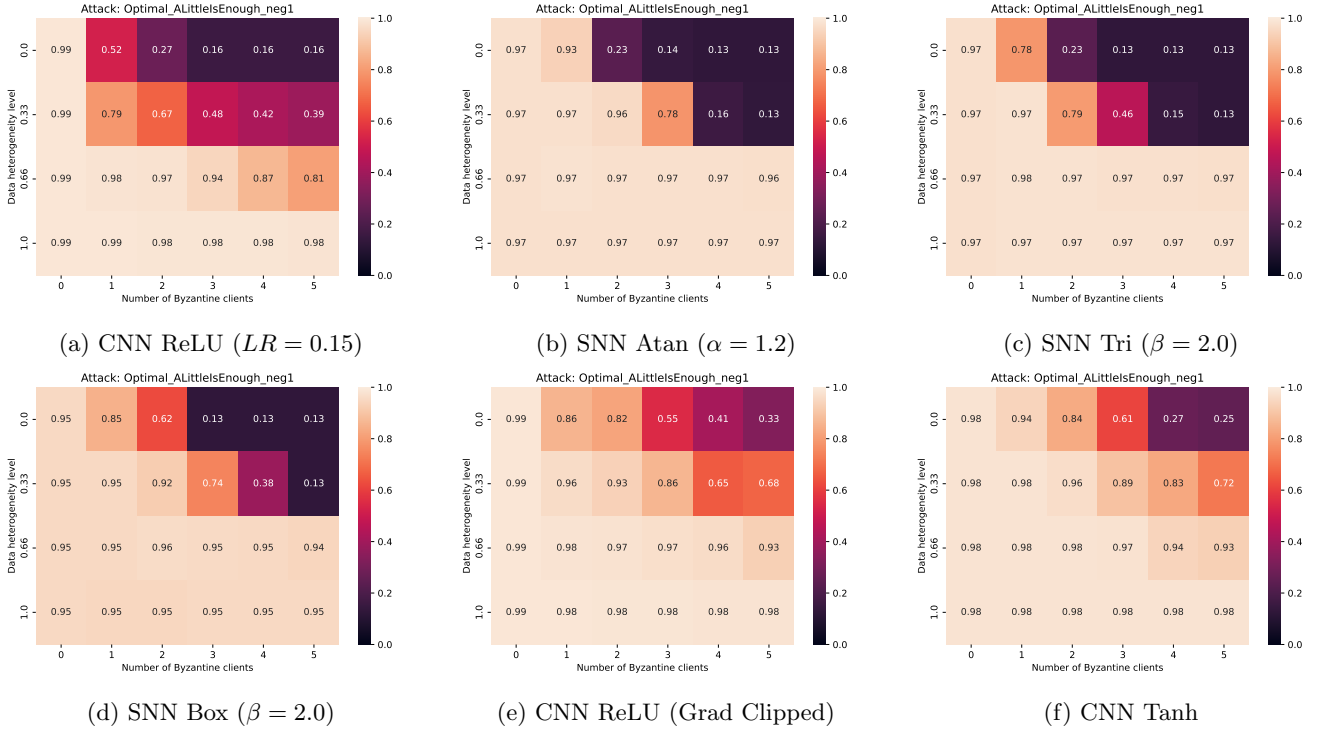


Figure 2: Best test accuracy under OptALIE.

3.2 Sign Flipping Attack

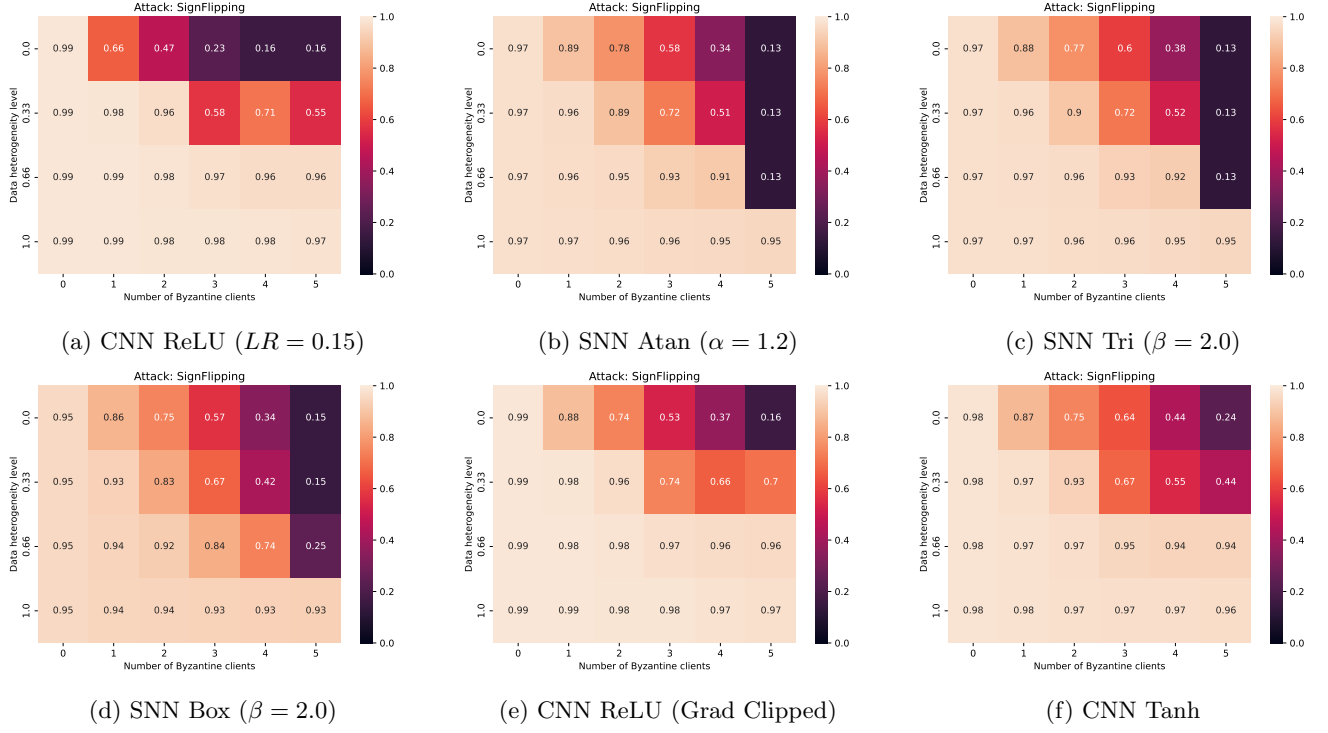


Figure 3: Best test accuracy under Sign Flipping.

3.3 Optimal IPM Attack

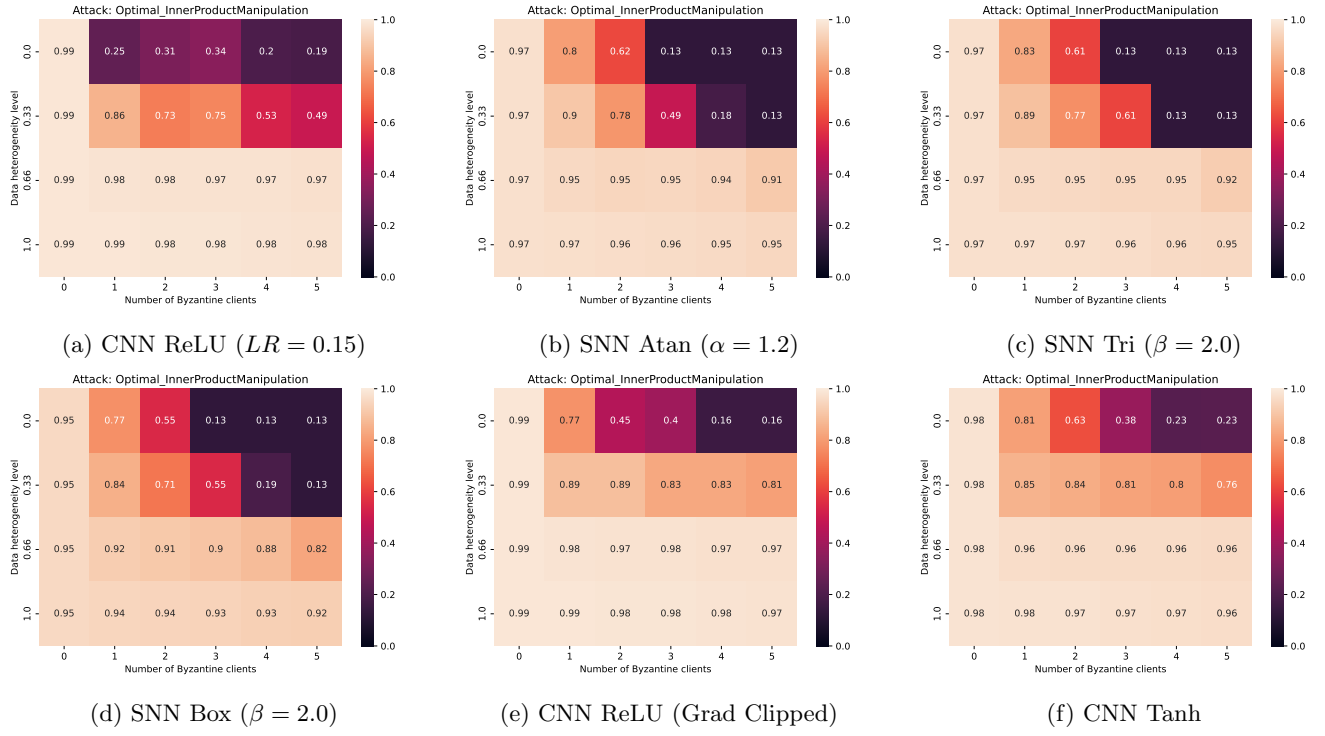


Figure 4: Best test accuracy under OptIPM.

3.4 Centered Clipping (CC)

3.4.1 CC under Worst-Case across all Attacks

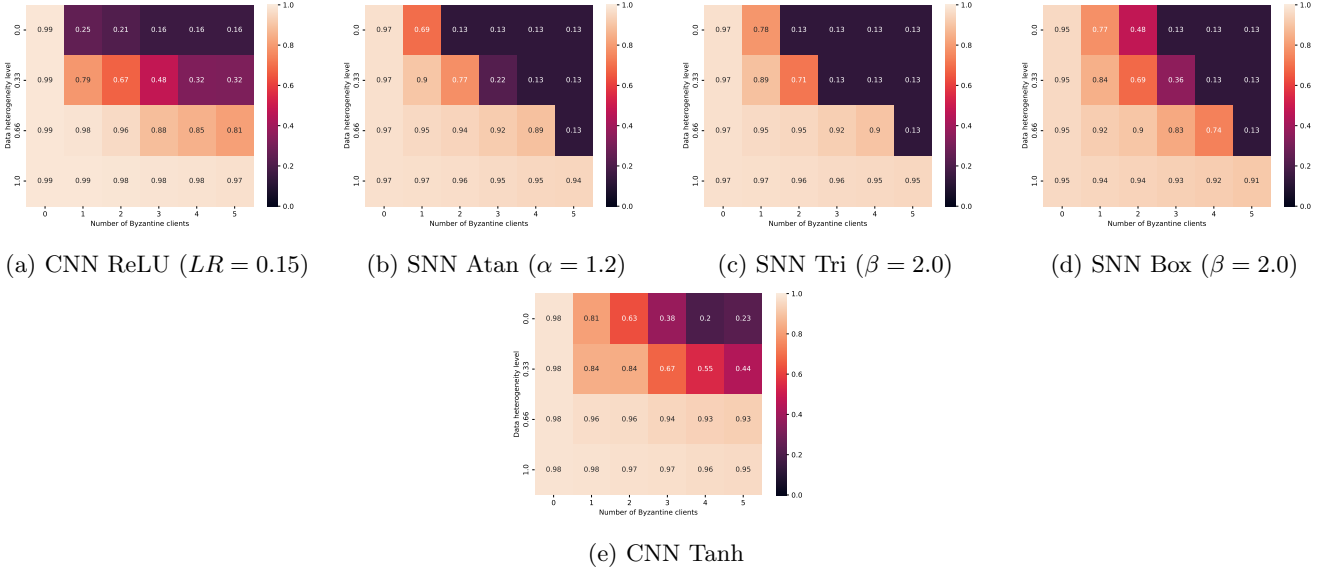
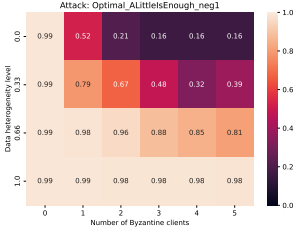
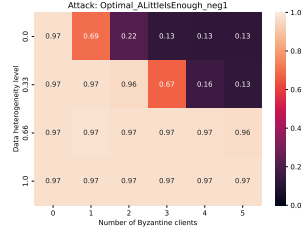


Figure 5: Centered Clipping under Worst-Case across all Attacks.

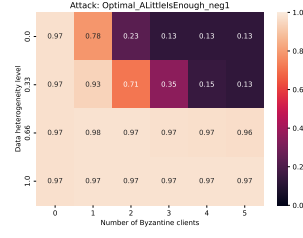
3.4.2 CC under Optimal ALIE



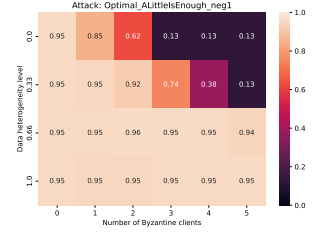
(a) CNN ReLU ($LR = 0.15$)



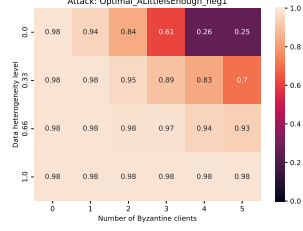
(b) SNN Atan ($\alpha = 1.2$)



(c) SNN Tri ($\beta = 2.0$)



(d) SNN Box ($\beta = 2.0$)



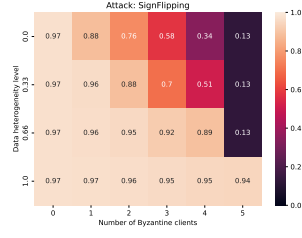
(e) CNN Tanh

Figure 6: Centered Clipping under Optimal ALIE.

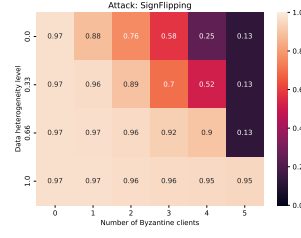
3.4.3 CC under Sign Flipping



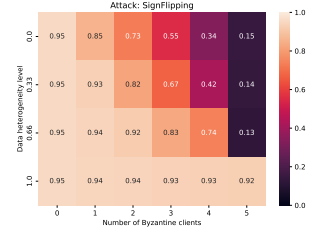
(a) CNN ReLU ($LR = 0.15$)



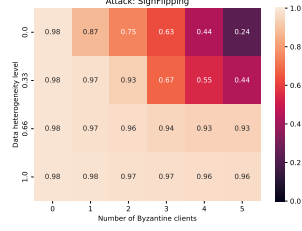
(b) SNN Atan ($\alpha = 1.2$)



(c) SNN Tri ($\beta = 2.0$)



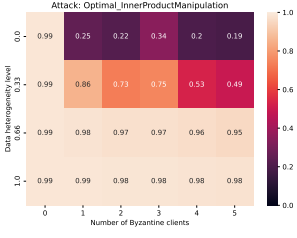
(d) SNN Box ($\beta = 2.0$)



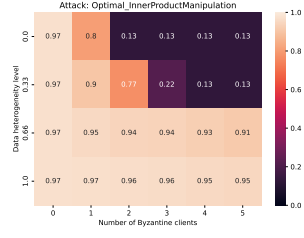
(e) CNN Tanh

Figure 7: Centered Clipping under Sign Flipping.

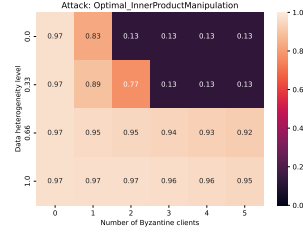
3.4.4 CC under Optimal IPM



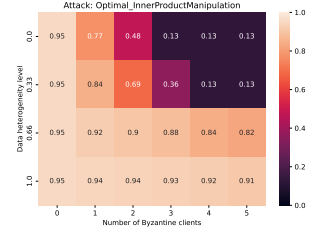
(a) CNN ReLU ($LR = 0.15$)



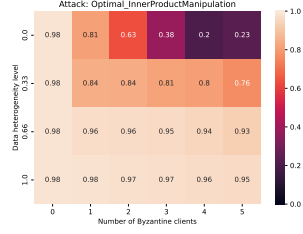
(b) SNN Atan ($\alpha = 1.2$)



(c) SNN Tri ($\beta = 2.0$)



(d) SNN Box ($\beta = 2.0$)



(e) CNN Tanh

Figure 8: Centered Clipping under Optimal IPM.

3.5 Geometric Median (GM)

3.5.1 GM under Worst-Case across all Attacks

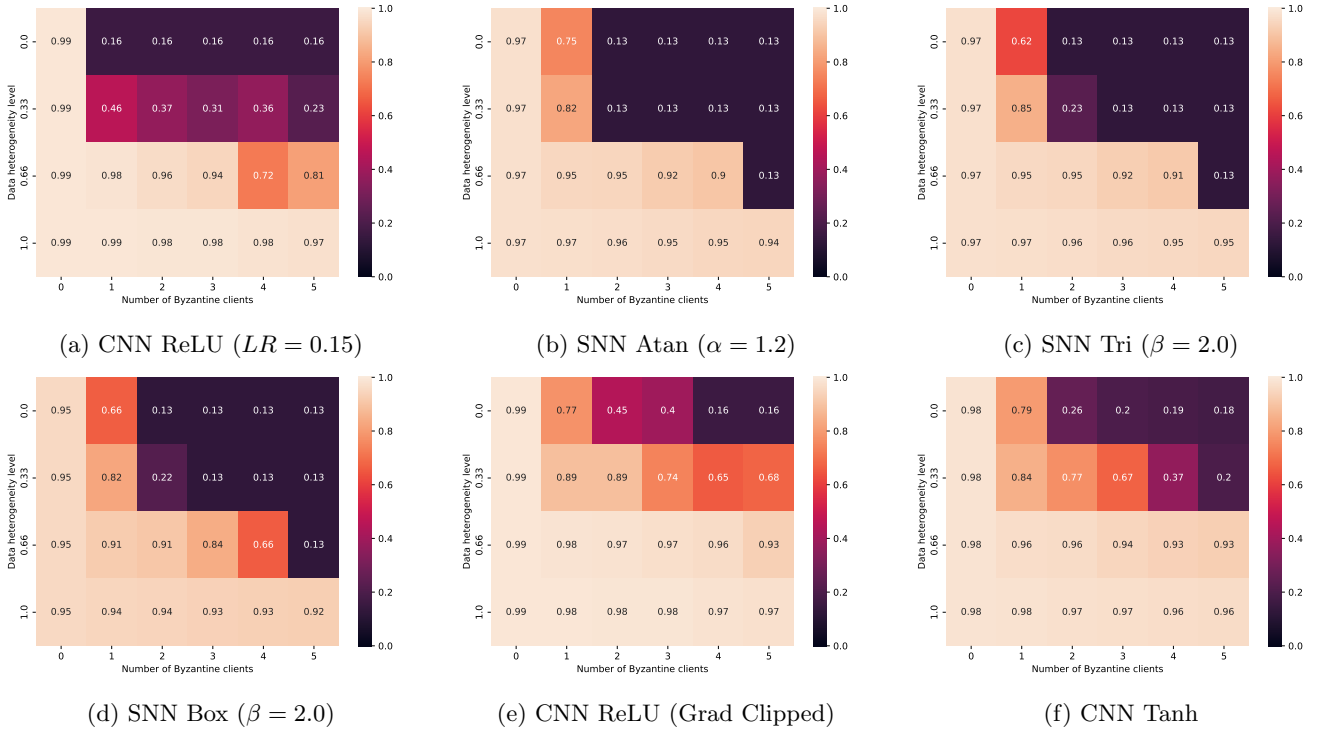


Figure 9: Geometric Median under Worst-Case across all Attacks.

3.5.2 GM under Optimal ALIE

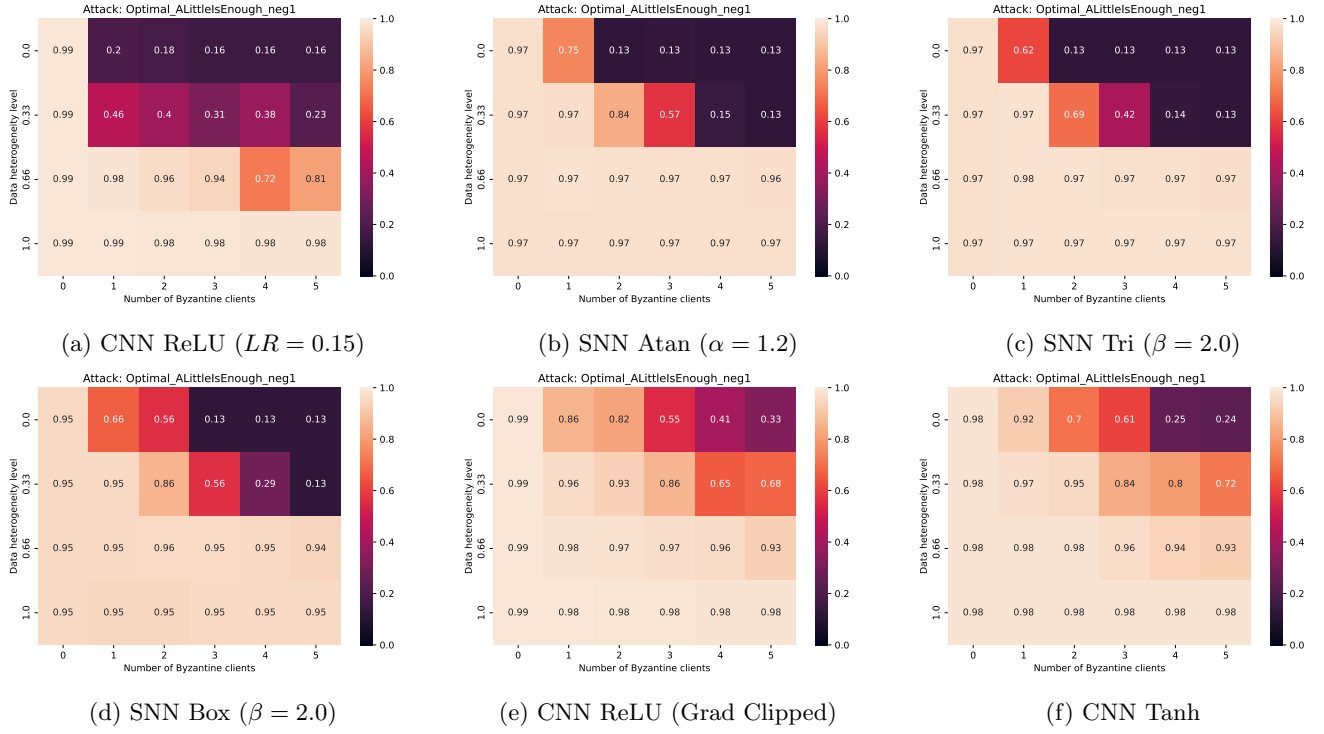


Figure 10: Geometric Median under Optimal ALIE.

3.5.3 GM under Sign Flipping

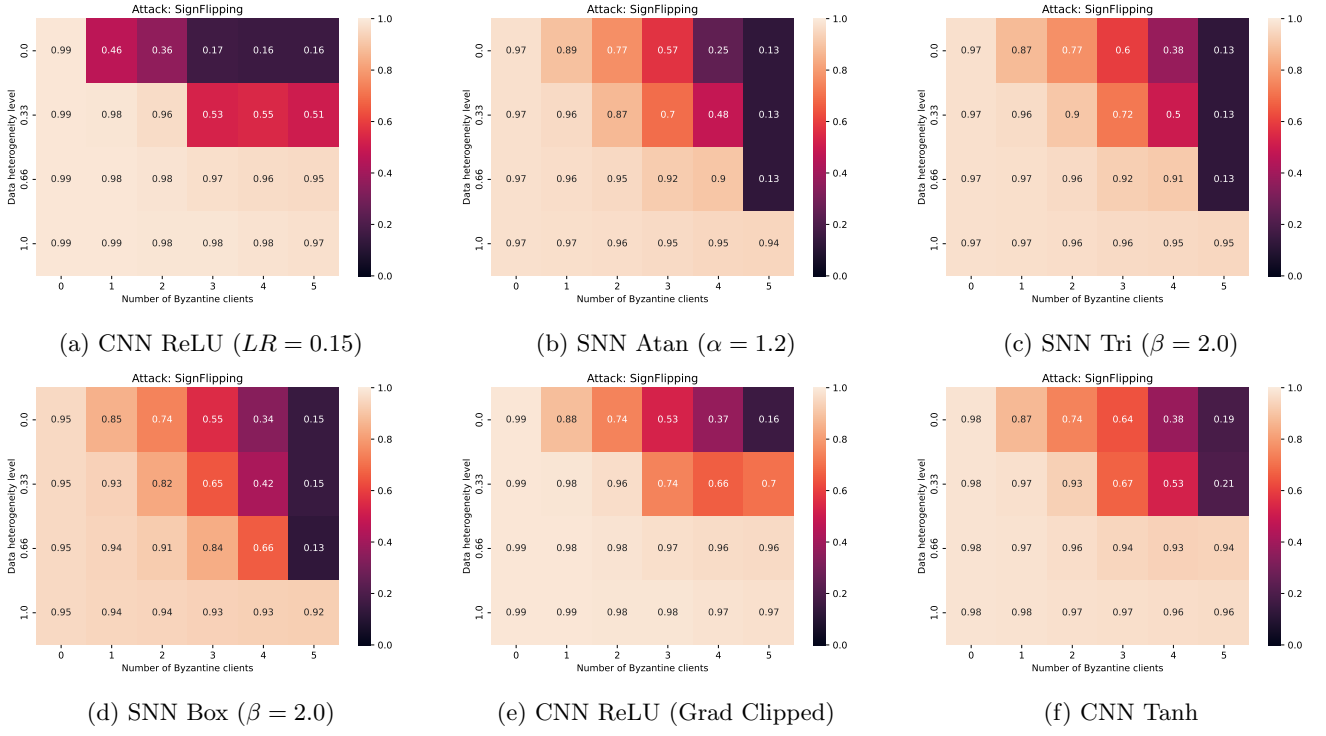


Figure 11: Geometric Median under Sign Flipping.

3.5.4 GM under Optimal IPM

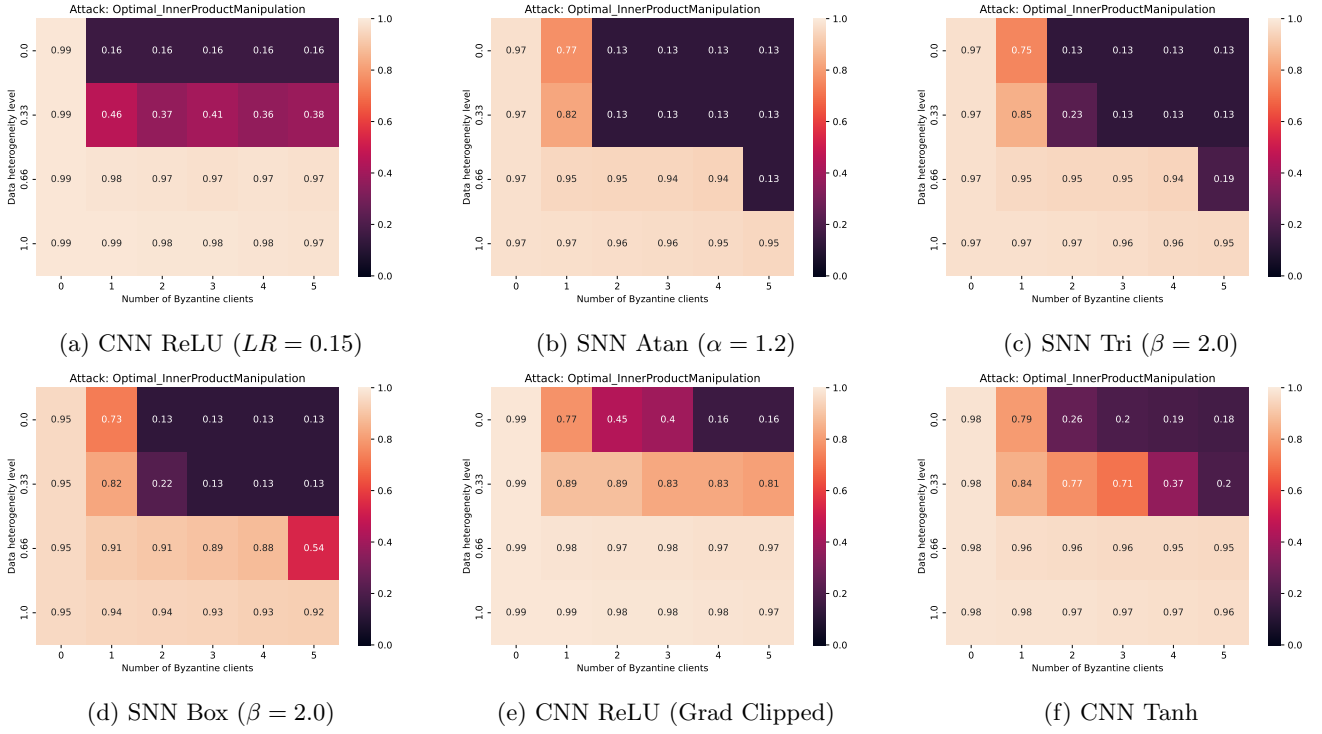


Figure 12: Geometric Median under Optimal IPM.

3.6 Multi-Krum (MK)

3.6.1 MK under Worst-Case across all Attacks

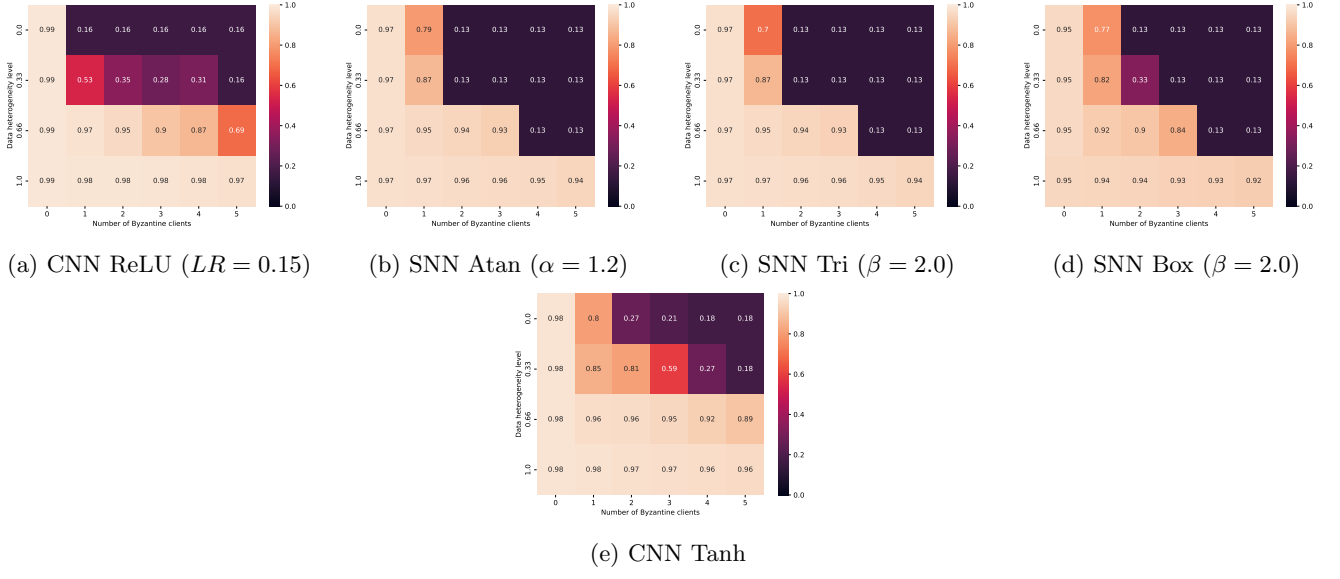


Figure 13: Multi-Krum under Worst-Case across all Attacks.

3.6.2 MK under Optimal ALIE

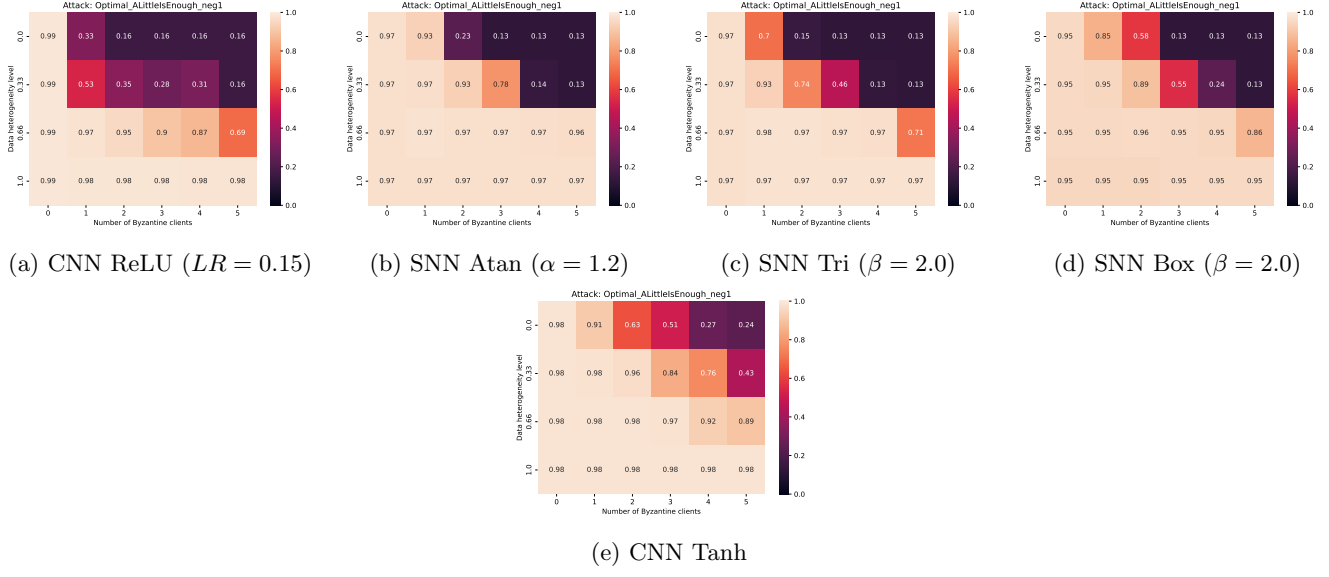
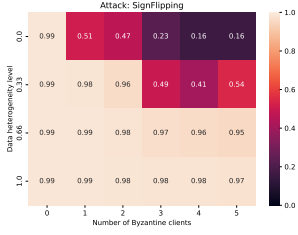
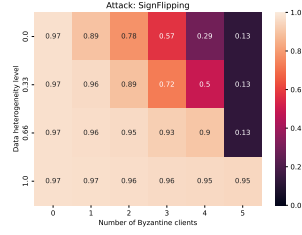


Figure 14: Multi-Krum under Optimal ALIE.

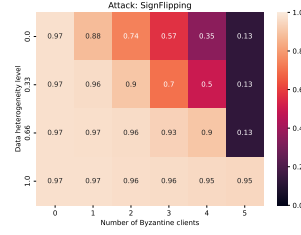
3.6.3 MK under Sign Flipping



(a) CNN ReLU ($LR = 0.15$)



(b) SNN Atan ($\alpha = 1.2$)



(c) SNN Tri ($\beta = 2.0$)



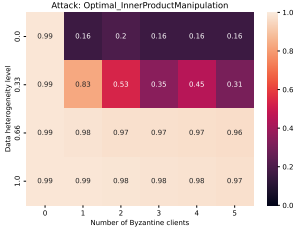
(d) SNN Box ($\beta = 2.0$)



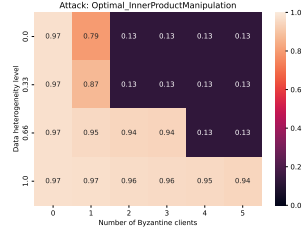
(e) CNN Tanh

Figure 15: Multi-Krum under Sign Flipping.

3.6.4 MK under Optimal IPM



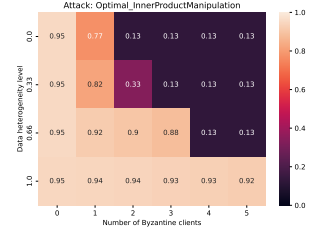
(a) CNN ReLU ($LR = 0.15$)



(b) SNN Atan ($\alpha = 1.2$)



(c) SNN Tri ($\beta = 2.0$)



(d) SNN Box ($\beta = 2.0$)



(e) CNN Tanh

Figure 16: Multi-Krum under Optimal IPM.

3.7 Trimmed Mean (TM)

3.7.1 TM under Worst-Case across all Attacks

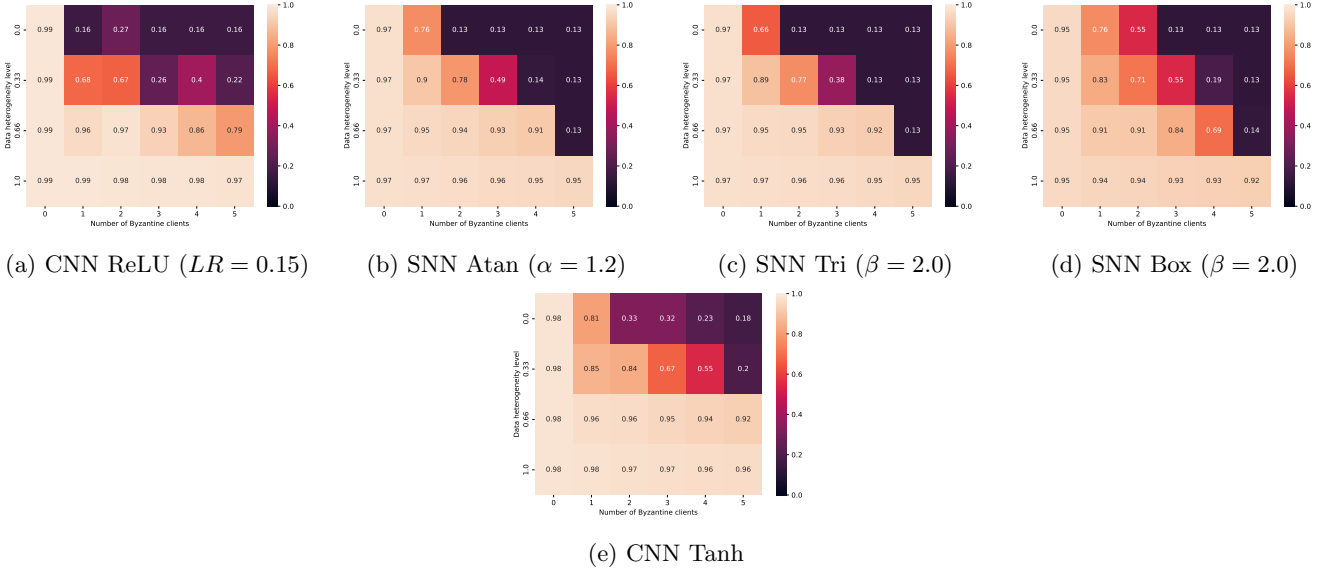


Figure 17: Trimmed Mean under Worst-Case across all Attacks.

3.7.2 TM under Optimal ALIE

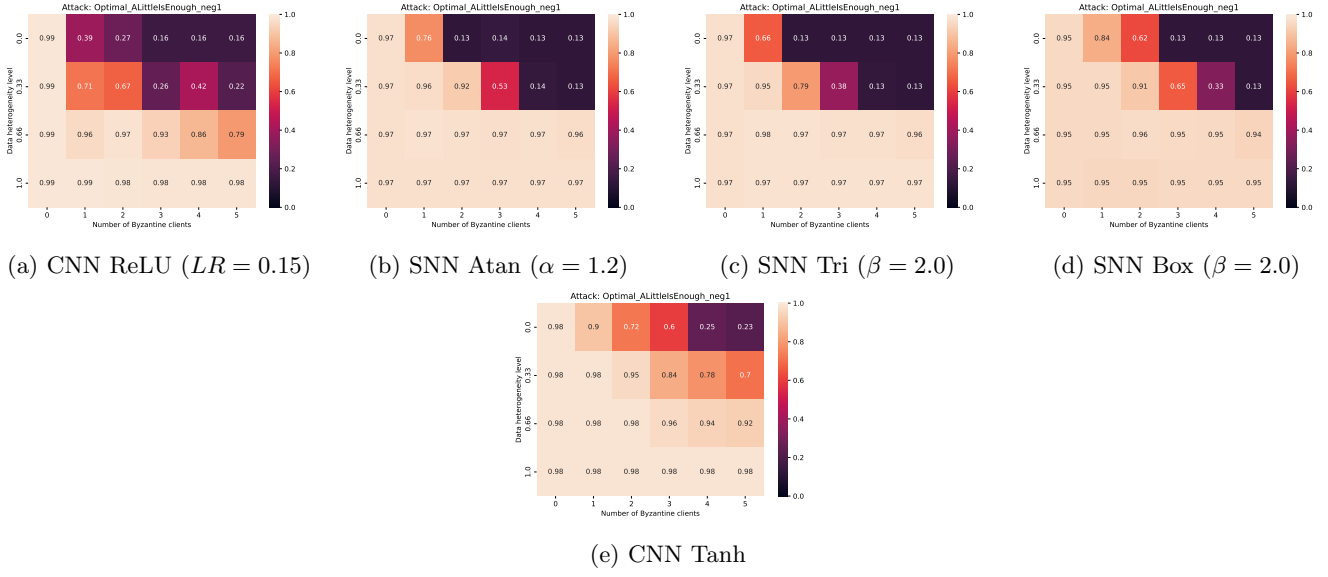
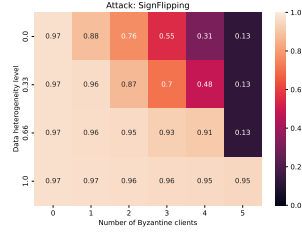


Figure 18: Trimmed Mean under Optimal ALIE.

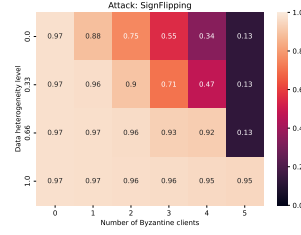
3.7.3 TM under Sign Flipping



(a) CNN ReLU ($LR = 0.15$)



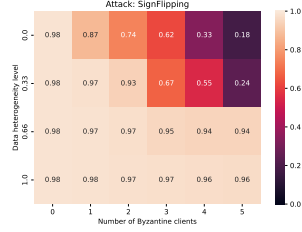
(b) SNN Atan ($\alpha = 1.2$)



(c) SNN Tri ($\beta = 2.0$)



(d) SNN Box ($\beta = 2.0$)



(e) CNN Tanh

Figure 19: Trimmed Mean under Sign Flipping.

3.7.4 TM under Optimal IPM

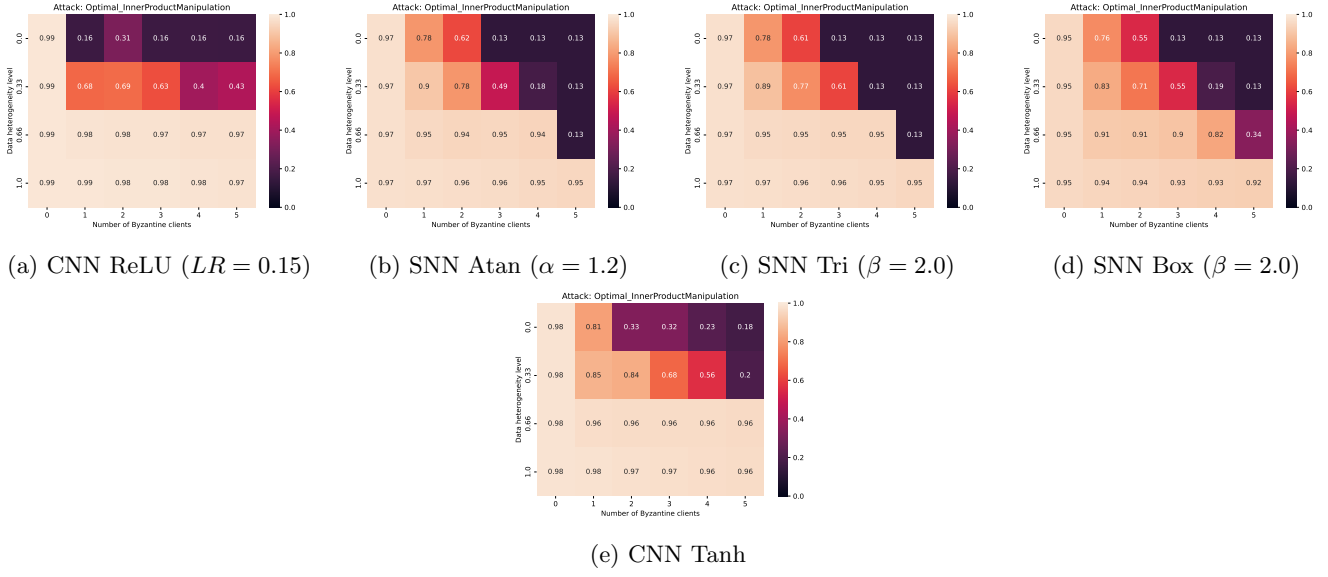


Figure 20: Trimmed Mean under Optimal IPM.